Hints: Methods of proof – 1

Homework

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Note. "Rationally independent α_i" is not enough for Kronecker's theorem. For example, $\sqrt{2}$ and $1+\sqrt{2}$ are linearly independent over $\mathbb{Q}$, but the points $(\{q\sqrt{2}\}, \{q(1+\sqrt{2})\})$ all lie on the diagonal. The correct hypothesis is that $1, \alpha_1, \dots, \alpha_k$ are linearly independent over $\mathbb{Q}$, and the hints use it.

Hint 1. Cut the unit cube $[0,1)^k$ into N^k small cubes of side $1/N$, and place the N^k+1 points $(\{q\alpha_1\}, \dots, \{q\alpha_k\})$, $q = 0, 1, \dots, N^k$, into them.

Hint 2. Two points with indices $a < b$ fall into the same small cube. Then $q = b - a \leqslant N^k$ satisfies $\|q\alpha_i\| < 1/N$ for every i.

Hint 3. For density, recurrence near 0 is not enough when $k \geqslant 2$: multiples of one small vector only fill a line. Use Weyl's criterion instead. For an integer vector $h \neq 0$, the number $h \cdot \alpha$ is not an integer (this is exactly where the independence of $1, \alpha_1, \dots, \alpha_k$ enters). So $\frac{1}{Q}\sum_{q=1}^{Q} e^{2\pi i q\, h\cdot\alpha}$ is a geometric sum bounded by $\frac{2}{Q|1-e^{2\pi i h\cdot\alpha}|} \to 0$.

Hint 4. Hence $\frac{1}{Q}\sum_{q\leqslant Q} P(q\alpha) \to \int_{[0,1]^k} P$ for every trigonometric polynomial P. Given a small open box B, take a continuous $0 \leqslant \varphi \leqslant \mathbf{1}_B$ with $\int \varphi > 0$ and approximate it uniformly by a trigonometric polynomial (Fejér or Stone–Weierstrass). The averages of $\varphi(q\alpha)$ have a positive limit, so some $q\alpha$ lies in B modulo $\mathbb{Z}^k$.

Problem 2

Note. The optimal constant d_n (the largest possible minimum distance of n points in a unit square) is not known in closed form for general n. It is the classical problem of packing equal circles in a square, solved only for small n. The hints cover the case $n = 5$, the order of magnitude $d_n \asymp n^{-1/2}$, and the true asymptotics.

Hint 1. For five points, cut the square into four squares of side $\frac{1}{2}$. Two points lie in the same small closed square, so their distance is at most its diagonal, $\sqrt{2}/2$. The bound is attained by the four corners together with the centre.

Hint 2. Upper bound for n points: if $m^2 < n$, cut the square into m^2 squares of side $1/m$. This gives $d_n \leqslant \sqrt{2}/m$, so $d_n \leqslant \sqrt{2}/(\lceil\sqrt{n}\rceil - 1) \sim \sqrt{2/n}$. Lower bound: choose n points of the $k \times k$ grid with spacing $1/(k-1)$, where $k = \lceil\sqrt{n}\rceil$. This gives $d_n \geqslant 1/(\lceil\sqrt{n}\rceil - 1) \geqslant 1/\sqrt{n}$.

Hint 3. A better upper bound comes from area instead of boxes. The disks of radius $d_n/2$ around the points are disjoint and lie in the square enlarged by $d_n/2$ on each side, so $n\pi d_n^2/4 \leqslant (1+d_n)^2$. This gives $d_n \leqslant (1+o(1))\, 2/\sqrt{\pi n}$.

Hint 4. The exact asymptotics is $d_n \sim \sqrt{2/(\sqrt{3}\,n)}$. The lower bound comes from a hexagonal (triangular lattice) arrangement. The upper bound needs Thue's theorem that the density of a packing of equal disks is at most $\pi/(2\sqrt{3})$. Boundary effects are $O(\sqrt{n})$ disks out of n.

Problem 3

Hint 1. 2^n begins with the digit string 2026 if and only if $2026 \cdot 10^j \leqslant 2^n < 2027 \cdot 10^j$ for some j. Take $\log_{10}$: this says that the fractional part $\{n \log_{10} 2\}$ lies in $[\log_{10} 2.026,\ \log_{10} 2.027)$.

Hint 2. The number $\log_{10} 2$ is irrational: $2^a = 10^b$ with $a \geqslant 1$ is impossible, because $5 \mid 10^b$ but $5 \nmid 2^a$. So the sequence $\{n \log_{10} 2\}$ is dense in $[0, 1]$ (pigeonhole, as in the class work), and it meets every interval of positive length.

Hint 3. The same argument works for any digit string D whose first digit is non-zero: use the interval $[\log_{10}(D/10^r), \log_{10}((D+1)/10^r))$, where $r+1$ is the number of digits of D. So exactly the strings that do not start with 0 occur.

Problem 4

Hint 1. Density theorem: for measurable $A \subseteq \mathbb{R}$, almost every $x \in A$ satisfies $m(A \cap I)/|I| \to 1$ as intervals $I \ni x$ shrink to x. It suffices to show that for each $\varepsilon > 0$ the set E of points of A where $\liminf m(A \cap I)/|I| < 1 - \varepsilon$ has measure 0.

Hint 2. Suppose $m^*(E) > 0$ (assume A is bounded). Take an open $U \supseteq A$ with $m(U) < m(A) + \eta$. Each $x \in E$ lies in arbitrarily small intervals $I \subseteq U$ with $m(I \setminus A) > \varepsilon |I|$. By the Vitali covering lemma, choose disjoint such intervals that cover E up to a null set. These intervals have total length at least $m^*(E)$, so $\varepsilon\, m^*(E) \leqslant \sum m(I \setminus A) \leqslant m(U \setminus A) < \eta$, which is a contradiction for small η.

Hint 3. Steinhaus: take a density point x of A and an interval I around x with $m(A \cap I) > \frac{3}{4}|I|$. For $|t| < |I|/2$, the sets $A \cap I$ and $(A \cap I) + t$ both lie in $I \cup (I + t)$, which has length $|I| + |t| < \frac{3}{2}|I|$. Their measures add up to more than $\frac{3}{2}|I|$, so they intersect, and $t \in A - A$.

Hint 4. Both proofs extend to $\mathbb{R}^n$: the density theorem holds there with balls or cubes, by the Vitali lemma in $\mathbb{R}^n$. The regularity proof from the class work ($K \subseteq A$ compact, $U \supseteq K$ open with $m(U) < 2m(K)$, and $K + t \subseteq U$ for small t) needs no covering lemma at all. It extends further, to every locally compact group with Haar measure, where density theorems can fail.

Problem 5

Note. (1) For odd n the first claim is false: $\{\frac{n+1}{2}, \ldots, n\}$ has $\frac{n+1}{2} > \frac{n}{2}$ elements, and none of them divides another (e.g. $\{2, 3\}$ for $n = 3$). The correct hypothesis is $|A| > \lceil n/2 \rceil$, which is the same as $|A| > n/2$ for even n. (2) As stated, the second claim holds as soon as $|A| \geqslant 2$, and the bound $2n/3$ plays no role. The hints prove both claims in these forms.

Hint 1. Use the odd parts as boxes, as in the class work. Writing $a = 2^s u$ with u odd, there are $\lceil n/2 \rceil$ possible values of $u \leqslant n$. Two elements with the same odd part divide one another.

Hint 2. For $A + A$: take any two elements $a < b$ of A and look at $a + a$, $a + b$, $b + b$.

Problem 6

Note. The first claim is false: the $n + 1$ vectors from the centre of a regular simplex to its vertices make pairwise obtuse angles. The correct statement is that any $n + 2$ vectors in $\mathbb{R}^n$ contain two with angle at most $\pi/2$, i.e. at most $n + 1$ vectors can be pairwise obtuse. This gives the second claim for $n \geqslant 2$, since $2^n \geqslant n + 2$. For $n = 1$ the vectors $1, -1$ are a counterexample. The hints prove the bound $n + 1$.

Hint 1. Pairwise obtuse means $\langle v_i, v_j \rangle < 0$ for all $i \neq j$. Induct on n. In $\mathbb{R}^1$ at most two non-zero numbers can have pairwise negative products.

Hint 2. Given pairwise obtuse $v_0, \ldots, v_m$ in $\mathbb{R}^n$, project $v_1, \ldots, v_m$ onto $v_0^\perp$: write $v_i = w_i + c_i v_0$, where $c_i = \langle v_i, v_0 \rangle / |v_0|^2 < 0$.

Hint 3. Then $\langle w_i, w_j \rangle = \langle v_i, v_j \rangle - c_i c_j |v_0|^2 < 0$. So $w_1, \ldots, w_m$ are pairwise obtuse (in particular non-zero) vectors in an $(n-1)$-dimensional space. By induction $m \leqslant n$, so there are at most $n + 1$ vectors. The simplex shows the bound is sharp.

Problem 7

Note. The inequality $X \geqslant m - \lambda\sigma$ only bounds the lower tail and cannot by itself push a random cut above its mean $m/2$. The hints prove the inequality and then get the excess $c\sqrt{m}$ from a lower bound on $\mathrm{E}|X - \mathrm{E}X|$, which is again a moment argument.

Hint 1. The inequality is Chebyshev: $P(X < m - \lambda\sigma) \leqslant P(|X - m| \geqslant \lambda\sigma) \leqslant \operatorname{Var} X/(\lambda\sigma)^2 \leqslant 1/\lambda^2$.

Hint 2. Put independent random signs $\varepsilon_v = \pm 1$ on the vertices and let X be the number of edges whose endpoints get different signs. Then $X = \frac{m}{2} - \frac{1}{2}Y$ with $Y = \sum_{uv \in E} \varepsilon_u \varepsilon_v$, $\mathrm{E}Y = 0$ and $\mathrm{E}Y^2 = m$, because the products $\varepsilon_u\varepsilon_v$ for distinct edges are orthogonal.

Hint 3. Bound $\mathrm{E}Y^4$. Expanding, only ordered 4-tuples of edges in which every vertex appears an even number of times contribute 1. These are the tuples of the form e, e, f, f (at most $3m^2$ of them) and the 4-cycles. A pair of disjoint opposite edges lies in at most two 4-cycles, so the 4-cycles contribute $O(m^2)$. Hence $\mathrm{E}Y^4 \leqslant Cm^2$.

Hint 4. By Hölder, $(\mathrm{E}Y^2)^3 \leqslant (\mathrm{E}|Y|)^2\,\mathrm{E}Y^4$, so $\mathrm{E}|Y| \geqslant m^{3/2}/(\sqrt{C}\,m) = c'\sqrt{m}$. Since $\mathrm{E}Y = 0$, the negative part satisfies $\mathrm{E}Y^- = \frac{1}{2}\mathrm{E}|Y| \geqslant \frac{c'}{2}\sqrt{m}$, so $Y \leqslant -\frac{c'}{2}\sqrt{m}$ for some choice of signs. The cut defined by those signs has at least $\frac{m}{2} + \frac{c'}{4}\sqrt{m}$ edges.

Problem 8

Hint 1. Infinite pigeonhole: if each of the finitely many colour classes were finite, their union would be finite.

Hint 2. Infinite Ramsey for a 2-colouring (or r-colouring) of the pairs of $\mathbb{N}$: pick x_1. By pigeonhole, infinitely many y have the same colour c_1 on $\{x_1, y\}$; call this infinite set S_1. Pick $x_2 \in S_1$ and repeat inside S_1. This gives $x_1, x_2, \ldots$ such that the colour of $\{x_i, x_j\}$ with $i < j$ depends only on i (it equals c_i).

Hint 3. Apply pigeonhole once more to the colours c_i: infinitely many i share a colour, and those x_i form a monochromatic infinite set.

Hint 4. Finite Ramsey from infinite Ramsey: suppose that for some k, every N has a colouring of K_N with no monochromatic K_k. These colourings form a finitely branching tree (restrict a colouring of K_{N+1} to K_N). By König's lemma (compactness of $\{0,1\}^{\binom{\mathbb{N}}{2}}$) it has an infinite branch, i.e. a colouring of all pairs of $\mathbb{N}$ with no monochromatic K_k. This contradicts the infinite theorem.

Problem 9

Hint 1. Case $k = 1$ (and the main lemma). Let $\delta > 0$ be the upper density and $L + 1 > 1/\delta$. Show that among any integers $n_0 < \cdots < n_L$, some difference $n_i - n_j$ lies in $A - A$. Consider the translates $A + n_i$ restricted to $[1, N]$, for N along a sequence with $|A \cap [1, N]| \geqslant (\delta - \varepsilon)N$.

Hint 2. These $L+1$ translates lie in $[1, N+n_L]$ (shift so that $n_0 \geqslant 0$) and have total size at least $(L+1)(\delta-\varepsilon)N > N + n_L$ for large N. So two of them meet: $a + n_i = a' + n_j$, i.e. $n_j - n_i \in A - A$, and it is non-zero.

Hint 3. General k: suppose that for every $d \geqslant 1$ some $j(d) \in \{1, \ldots, k\}$ has $j(d)\,d \notin A - A$. Colour each pair $\{x < y\}$ of positive integers with the colour $j(y - x)$. By (finite) Ramsey's theorem with k colours, there are $L + 1$ integers $x_0 < \cdots < x_L$ all of whose pairs have the same colour j.

Hint 4. Then no difference of the $L + 1$ integers $jx_0, \ldots, jx_L$ lies in $A - A$, which contradicts the lemma of Hints 1–2. So some d has $d, 2d, \ldots, kd \in A - A$. The general case needs exactly this extra Ramsey step.

Problem 10

Hint 1. Erdős–Szekeres: label a_i with the pair (ℓ_i, r_i), the lengths of the longest increasing and longest decreasing subsequences ending at a_i. Distinct indices get distinct pairs: if $i < j$, then a_j extends one of the two sequences ending at a_i. With all $\ell_i, r_i \leqslant n$ there are only n^2 possible pairs.

Hint 2. Sharpness: take n blocks, each block a decreasing run of n numbers, with every block lying entirely above the previous one, e.g. $n, n-1, \ldots, 1,\ 2n, \ldots, n+1,\ \ldots$. An increasing subsequence takes at most one term from each block, and a decreasing one stays inside a single block.

Hint 3. Random permutation of $\{1, \ldots, N\}$: by Erdős–Szekeres, $\max(\mathrm{LIS}, \mathrm{LDS}) \geqslant \sqrt{N}$ always. Reversing the permutation swaps LIS and LDS, so $P(\mathrm{LIS} \geqslant \sqrt{N}) \geqslant \frac{1}{2}$.

Hint 4. For the matching upper bound (first moment): the expected number of increasing subsequences of length ℓ is $\binom{N}{\ell}/\ell! \leqslant (e^2 N/\ell^2)^\ell$, which tends to 0 for $\ell \geqslant (1+\varepsilon)e\sqrt{N}$. So the LIS of a random permutation is of order $\sqrt{N}$ with high probability.